

Development Guide

Prerequisites

- JDK 21
- Maven 3.9+
- Node.js + Angular CLI (frontend)
- PostgreSQL with DB `bankone` and user `bankone_user`
- Optional: Open Liberty under `~/tools/wlp` (server name `bankone`)

Backend

```
cd "BankOne/BankOne-BackEnd"
# Embedded
mvn spring-boot:run
# Liberty (recommended for this project)
./scripts/redeploy-liberty.sh
```

Key config: `BankOne-BackEnd/src/main/resources/application.properties`

Property	Purpose
<code>spring.datasource.*</code>	PostgreSQL connection
<code>spring.jpa.hibernate.ddl-auto=update</code>	Schema sync
<code>spring.sql.init.mode=always</code>	Runs <code>schema.sql</code>
<code>server.port=8080</code>	Embedded only
<code>jwt.secret</code> / <code>jwt.expiration</code>	JWT signing

After Liberty redeploy, smoke login expects `admin` / `Admin@123`.

Frontend

```
cd "BankOne/BankOne-Frontend"
npm install # first time
ng serve
```

Set API host in `src/app/core/config/api-config.ts` (currently points at Liberty / LAN host on port `9080`).

Coding conventions

Backend

- Package by domain: `com.bankone.{module}.{controller|service|repository|entity|dto|...}`
- Prefer service interface + `*Impl`

- Search via JPA `Specification` classes
- Map API responses through DTOs for accounts/policies; customers currently return entity (known smell)
- Security roles: `ROLE_` prefix handled by Spring; matchers use `ADMIN` , `EMPLOYEE` , `MANAGER`

Frontend

- Feature folders under `src/app/features/`
- Shared HTTP in `src/app/core/services/`
- Models in `src/app/core/models/`
- Bearer token via `auth.interceptor.ts`
- Dialogs as standalone components (e.g. `opening-deposit-dialog/`)

Teach-while-building

Project rule (`.cursor/rules/teach-while-building.mdc`): unless the user says to apply code, explain snippets for IntelliJ paste. Documentation sync still applies when code lands.

Documentation sync checklist (every change)

1. Update module doc under `docs/MODULES/`
2. Update API / schema / sequences / call flow if surfaces changed
3. Append `docs/CHANGELOG.md`
4. Refresh Mermaid diagrams when structure changes

Common pitfalls

Symptom	Likely cause
405 on new GET endpoint	Stale Liberty WAR — redeploy
Opening CURRENT fails on deposit	Policy requires min opening (seed <code>CURRENT/INR = 5000</code>)
CORS errors from phone/LAN	Firewall / AP isolation; CORS already allows <code>192.168.0.4</code>
Roles missing after login	Roles not in JWT — check <code>user_roles</code> and filter reload

Suggested next build slices

1. Transaction ledger foundation
2. Account detail + debit/credit history
3. Withdraw
4. Account list filters / polish
5. Status transition rules
6. Maker-checker